

Exercise 1: Set Up Your Data Model

Interfaces 102: Building Dynamic Interfaces

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Introduction

How Can I Practice?

You should practice in Appian Community Edition. Keep in mind that this is a community environment not suitable for production workloads or sensitive information.

Appian Version

Appian Community Edition is on the latest Appian version. If you are following the exercises from a previous Appian version, go to academy.appian.com to download the latest version.

Naming Conventions

When you register for Appian Community Edition, you gain access to a workspace, which is your personal area within the community environment. When your workspace has been assigned, you will receive a confirmation email with a workspace ID. Pre-loaded apps in your workspace include the workspace ID.

Use this workspace ID when creating new applications and objects. Otherwise, you could run into naming conflicts with objects created by other users. In the example below, the application prefix for the Acme Auto Solution app is W0000AS. So, the workspace ID is W0000, which would be included in any new app's prefix.

Save Often

Appian does not automatically save updates, so save your objects frequently.

Additional Resources

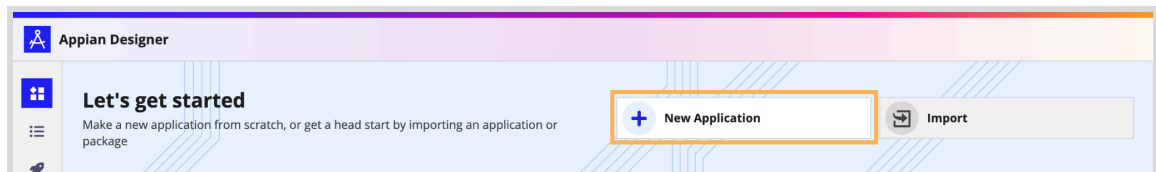
Appian provides a number of training resources for Appian developers. The following resources are particularly popular with our learners:

- [Academy Online](#) - Appian's online courses provide useful survey courses, step-by-step tutorials, and practice exercises. Explore these resources at your own pace. Survey courses will help you develop a better grasp of the topics you need to learn. Video and print tutorials will help you with getting hands-on experience with Appian.
- [Community Discussions for New Users](#) - Check out the **New to Appian** thread in Community. Join our community of experts to ask questions and find answers from past discussions.
- [Appian Documentation](#) - Appian's product documentation will provide you with an overview of key Appian features, newest release information, additional tutorials, and helpful patterns and recipes to implement in your app.

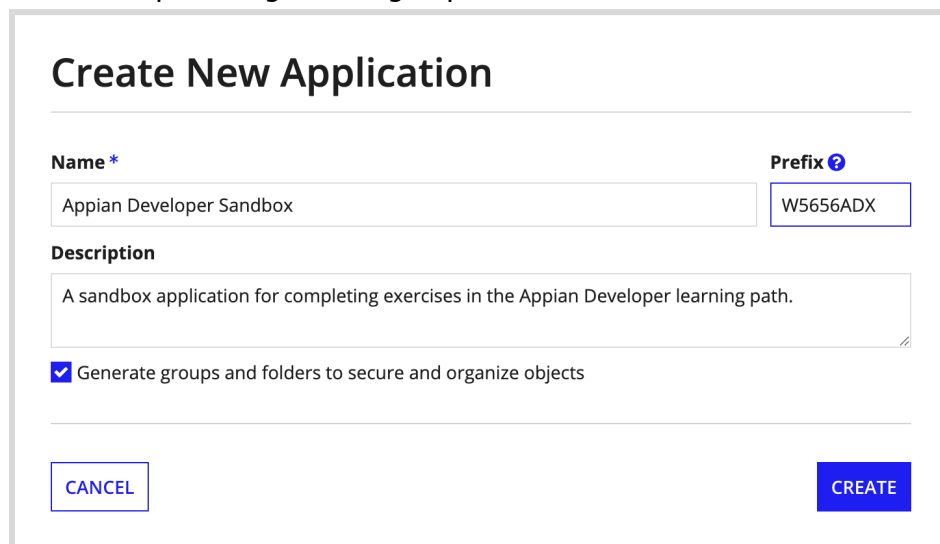
Create an Application

You may have been asked to create a sandbox application in your **Appian Community Edition** workspace in another course. If so, feel free to skip to **Create the Record Types**. Otherwise, you will need to create an application to hold the objects used in the exercises for this course. Follow the steps below:

1. Navigate to **Appian Designer** in your Appian Community Edition workspace.
2. Click **New Application**.



3. In the dialog, enter the following:
 - **Name:** Enter Appian Developer Sandbox
 - **Prefix:** Enter W#ADX (W# should be replaced with your workspace identifier)
 - Leave the option to generate groups and folders selected.

The screenshot shows the 'Create New Application' dialog box. It has a title bar 'Create New Application'. Inside, there are two input fields: 'Name' with the value 'Appian Developer Sandbox' and 'Prefix' with the value 'W5656ADX'. Below these is a 'Description' text area containing 'A sandbox application for completing exercises in the Appian Developer learning path.' At the bottom, there is a checkbox labeled 'Generate groups and folders to secure and organize objects' which is checked. At the very bottom, there are two buttons: 'CANCEL' and 'CREATE'.

4. Click **CREATE**.
5. Leave the default security settings, and click **SAVE**.

Create the Record Types

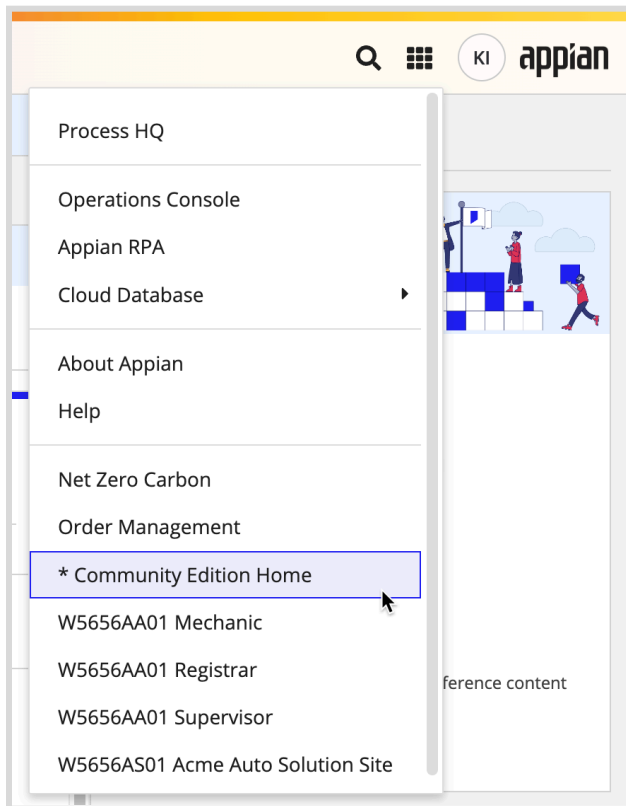
Throughout this course, you'll gradually add functionality to an interface that allows automobile maintenance technicians to create an estimate for an existing maintenance request by adding parts and costs.

This process will take advantage of a few record types, one of which is available in the **Acme Automobile Reference Application** (available from the App Catalog in Appian Community Edition), the others you will have to create yourself.

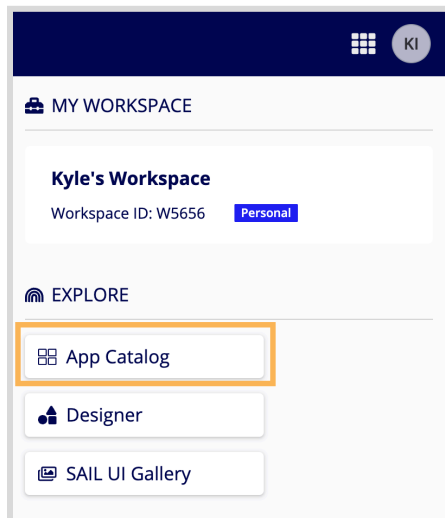
Deploy the Acme Automobile Applications

If you have not already done so, deploy the Acme Automobile Applications by following these steps:

1. In your Appian Community Edition workspace, navigate to the **Community Edition Home** site from the navigation menu.



2. In the pane on the right, select the **App Catalog**.



- From the available options, select **Acme Automobile Applications**.

APPLICATIONS Deployment Tokens: 1

Acme Automobile Applications
Tutorial Beginner
Manage vehicle additions and maintenance within a commercial fleet while practicing with courses on Academy Online.

Order Management
Tutorial Beginner
Get hands-on practice with Process HQ using this sample order management use case.

Advanced Expressions
Tutorial Intermediate
Get hands-on practice with indexing values, applying best practices for expression writing, using looping functions, and more!

AI Skills
Tutorial Beginner
Automate business tasks with document classification, document extraction, and email classification AI skills to streamline workflows effortlessly.

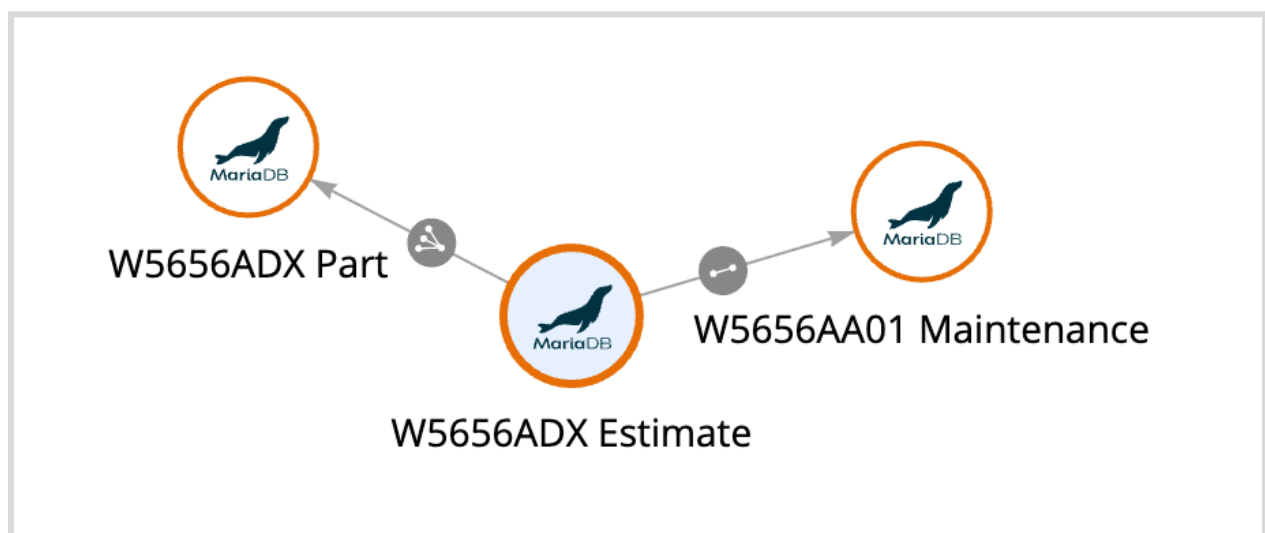
Acme IT Management
Tutorial Beginner
In need of a system to oversee all IT assets organization-wide, plan and develop an application to track equipment and handle asset requests.

Wyndham Power Outage
Tutorial Beginner
Empower Wyndham Power customers to report outages with a user-friendly portal. This app includes built-in ops for seamless monitoring.

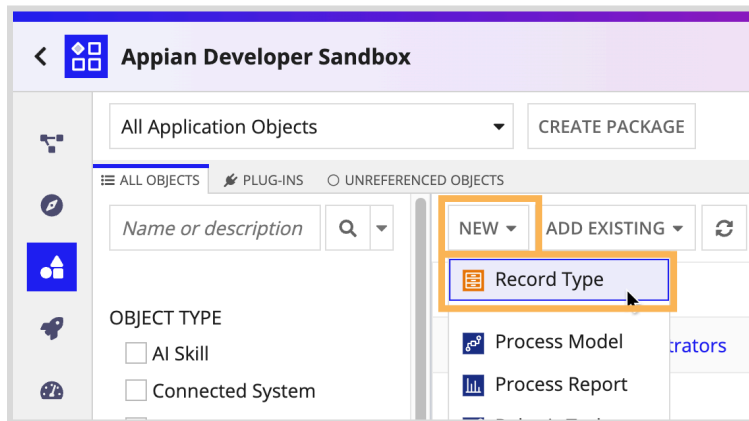
- Click **DEPLOY TO WORKSPACE**. This may take several minutes to complete. You can start setting up the other record types in the meantime.

Create the Estimate Record Type

The data for the maintenance estimate use case is centered around the **W#ADX Estimate** record type. Each estimate is related to a single maintenance request (stored in the **W#AA Maintenance** record type) and multiple parts (in the **W#ADX Part** record type). You will create both the **W#ADX Estimate** and **W#ADX Part** record types in this exercise.



1. In the **Appian Developer Sandbox** application, in the **Build** view, select **New**, then **Record Type**.



2. In the **Create Record Type** dialog, enter the following:
 - **Name:** Enter **W#ADX Estimate** (Replace W# with your workspace identifier)
 - **Description:** Enter **Estimate for vehicle maintenance**.

Create Record Type

Name *

W5656ADX Estimate

Display Name (Plural) ? *

Estimates

Description ?

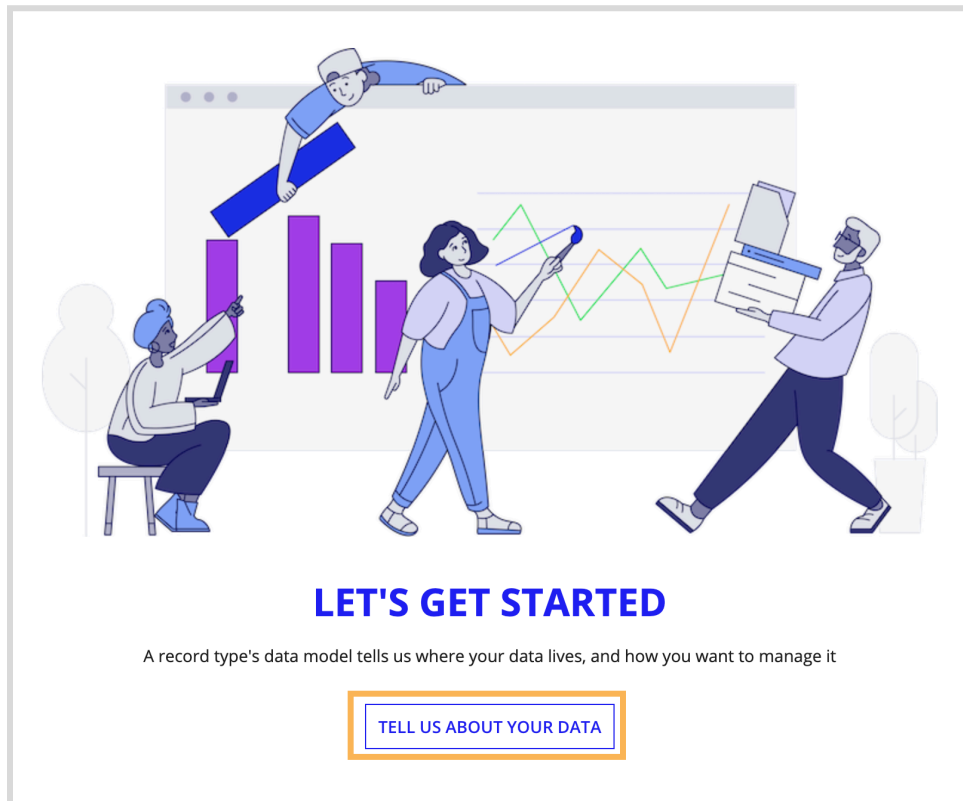
Estimate for vehicle maintenance

CANCEL

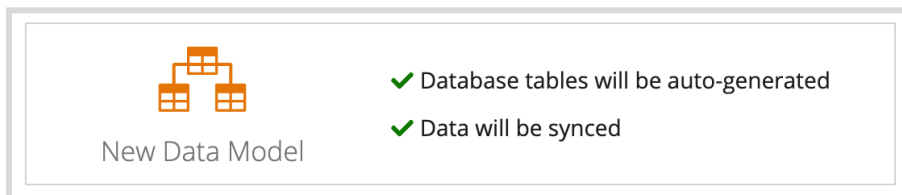
CREATE

3. Click **CREATE**.
4. In the **Review Record Type Security** dialog, leave the default security settings and click **SAVE**.

5. On the record type **Data Model** page, click **TELL US ABOUT YOUR DATA**.



6. Select the option **New Data Model** to start from scratch, then click **NEXT**.



7. Select the scheme that corresponds to your workspace identifier, then click **NEXT**.

Data Source ?

W5656 Data Source (Connected System)

Select Schema *

W5656

8. Under **Fields**, remove all fields except for **id**, then click **NEXT**.

Fields [NEW FIELD](#) [NEW CHOICE LIST](#)

 id Number (Integ... 

9. Under Relationships, click **NEW RELATIONSHIP**.

10. In the **Related Record Type** field, select the **W#AA Maintenance** record type. This record type is part of the Acme Automobile Reference Application you deployed earlier.

Then, click **NEXT**.

Add Relationship to W5656ADX Estimate

Related Record Type

 W5656AA01 Maintenance ✕

CANCELNEXT

11. Select the relationship type **One to One**. This will automatically create a new field in the **W#ADX Estimate** record type called **maintenanceId**. Then, click **ADD**.

Add Relationship to W5656ADX Estimate

Relationship Name *

maintenance

This will be used to reference the relationship and help access the related data. Choose a name that is descriptive and unique to this relationship.

Relationship Type

One to Many

W5656ADX Estimate



W5656AA01 Maintenance

Many to One

W5656ADX Estimate



W5656AA01 Maintenance

One to One

W5656ADX Estimate



W5656AA01 Maintenance

W5656ADX Estimate *

maintenanceld

Select common record fields for this relationship

 This unique common field will be automatically created for you
[Select an existing field instead](#)

W5656AA01 Maintenance *

id - Number (Integer)

Write and Delete Related Records *

☐ Write or delete Maintenance when modifying Estimates

☒ Do not write or delete Maintenance when modifying Estimates

BACKCANCELADD

12. Click **NEXT**.

13. On the next screen, review the record fields and relationships, then click **SAVE CHANGES**.

14. Finally, click **CLOSE**.

Create the Part Record Type

Next, you will create the **Part** record type. This record type will be used to list parts needed to perform vehicle maintenance, as well as their quantity and cost to include in the maintenance estimate.

1. In the **Build** view, select **New**, then **Record Type**.
2. In the **Create Record Type** dialog, enter the following:
 - **Name:** Enter `W#ADX Part` (Replace `W#` with your workspace identifier)
 - **Description:** Enter `Parts related to a vehicle maintenance estimate`.
3. Click **CREATE**.
4. In the **Review Record Type Security** dialog, leave the default security settings and click **SAVE**.
5. On the record type **Data Model** page, click **TELL US ABOUT YOUR DATA**.
6. Select the option **New Data Model** to start from scratch, then click **NEXT**.
7. Select the scheme that corresponds to your workspace identifier, then click **NEXT**.
8. Under **Fields**, configure the following fields:
 - **Field:** `id` **Type:** Number (Integer)
 - **Field:** `name` **Type:** Text
 - **Field:** `unitCost` **Type:** Number (Decimal)

- **Field:** quantity **Type:** Number (Integer)

+	id	Number (Integer)	🔗	+	↑	↓	×
+	name	Text		+	↑	↓	×
+	unitCost	Number (Decimal)		+	↑	↓	×
+	quantity	Number (Integer)		+	↑	↓	×

9. Click **NEXT**.
10. Under Relationships, click **NEW RELATIONSHIP**.
11. In the **Related Record Type** field, select the **W#ADX Estimate** record type. Then, click **NEXT**.

Add Relationship to W5656ADX Part

Related Record Type

W5656ADX Estimate

CANCEL NEXT

12. Leave the relationship type **Many to One** selected. This will automatically create a new field in the **W#ADX Part** record type called **estimateId**. Then, click **ADD**.

Add Relationship to W5656ADX Part

Relationship Name *

estimate

This will be used to reference the relationship and help access the related data. Choose a name that is descriptive and unique to this relationship.

Relationship Type

One to Many

W5656ADX Part

W5656ADX Estimate

Many to One

W5656ADX Part

W5656ADX Estimate

One to One

W5656ADX Part

W5656ADX Estimate

W5656ADX Part *

estimateId

W5656ADX Estimate ? *

id - Number (Integer)

Select common record fields for this relationship

✦ This common field will be automatically created for you

[Select an existing field instead](#)

Write and Delete Related Records ? *

☐ Write or delete Estimates when modifying Parts
 ☒ Do not write or delete Estimates when modifying Parts

BACK CANCEL ADD

13. Click **NEXT**.
14. On the next screen, review the record fields and relationships, then click **SAVE CHANGES**.

Update the Estimate Record Type

When you created the Part record type, you established a many-to-one relationship with the Estimate record type. Now you need to update the Estimate record type to add the reciprocal, one-to-many relationship there.

1. Open the **W#ADX Estimate** record type.
2. On the right, under **Relationships**, click **ADD RELATIONSHIP**.

SOURCE

Type Database

Name W5656 Data Source

Schema W5656

Table W5656_ADX_ESTIMATE

RELATIONSHIPS

ADD RELATIONSHIP

← maintenance

3. In the **Related Record Type** field, select the **W#ADX Part** record type. Then, click **NEXT**.

Add Relationship to W5656ADX Estimate

Related Record Type

W5656ADX Part

CANCEL NEXT

4. Leave the relationship type **One to Many** selected. Under **W#ADX Estimate**, select the field **id**. Under **W#ADX Part**, select the field **estimateId**. Then, click **ADD**.

Add Relationship to W5656ADX Estimate

Relationship Name *

This will be used to reference the relationship and help access the related data. Choose a name that is descriptive and unique to this relationship.

Relationship Type

One to Many
W5656ADX Estimate

W5656ADX Part

Many to One
W5656ADX Estimate

W5656ADX Part

One to One
W5656ADX Estimate

W5656ADX Part

W5656ADX Estimate ? *

W5656ADX Part ? *

Select common record fields for this relationship

Write and Delete Related Records ? *

☒ Write or delete Parts when modifying Estimates

☐ Do not write or delete Parts when modifying Estimates

Preview

With the record types and relationships configured, you are now prepared to complete the remaining exercises in this course.